

Lab – 7

Operator overloading problems

1. Create a class Time with three private variables `int h,m,s`; Create a function to overload '+' operator to add two time variables.

```
int main(){  
    Time t1(5,15,34),t2(9,53,58),t3;  
    t3 = t1 + t2; t3.show();  
}
```

2. Define a class STRING and overload == to compare two strings and + operator for concatenation of two strings.
3. Write a program in C++ to create a class matrix and overload * operator using friend function to multiply two matrices.
4. Overload '[]' to check array index out of bounds problem at run time.
5. Overload '()' to input arbitrary number of input arguments for an object.
6. Write a program in C++ to overload input operator (>>) and output operator (<<).
7. Write a program to convert basic data type (float) to user defined data type (object).

```
class Test {  
private: //....  
public:  
    Test ( data_type) { // conversion code }  
};
```

8. Write a program to convert UDT to basic data type (float)

```
class Test{  
public:  
    operator data_type() { //Conversion code }  
};
```

9. How will you convert one UDT to another UDT. For example conversion of polar to cartesian system.

```
Polar p(10,5);  
Cartesian c = p;  
c.show();
```